PRD-03 · Spoken interviews

Status: Shipped (v0.1.2) · **Updated:** 2026-10-30 · **Design:** DD-03 Voice pipeline

1. Summary

Real interviews are spoken. Onsite lets the candidate talk to the interviewer and hear it talk back, with all speech processed **on their own computer** by VoiceStudio (a separate local app) and a Whisper sidecar. Typing is always available, as a choice or as a fallback.

2. Why it matters

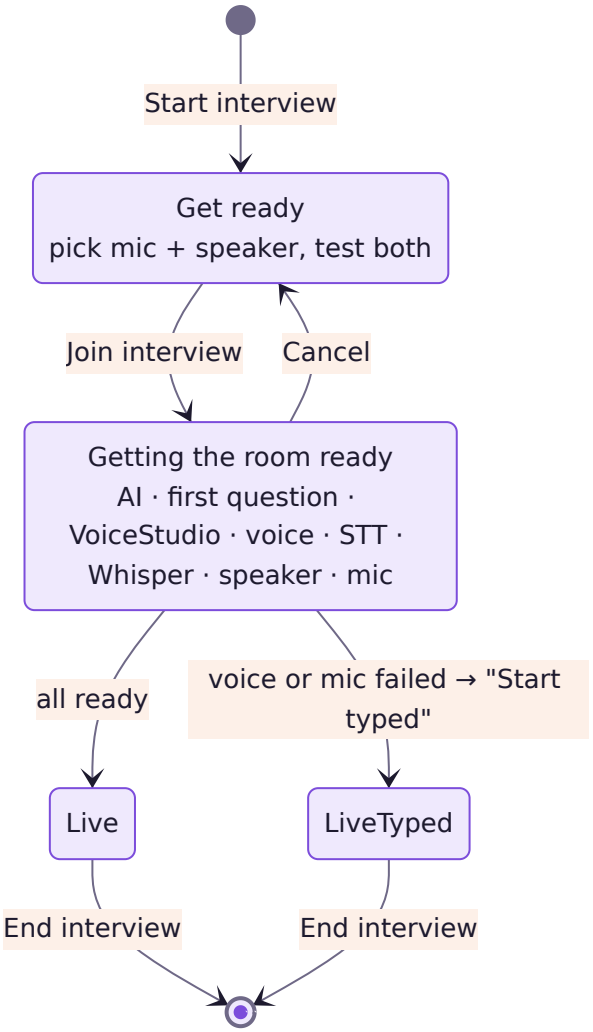
- Thinking out loud is the skill being tested. Typing hides rambling, filler and long silences.
- Interviewers interrupt and follow up mid-thought. Text chat can't train that.
- The first version (a browser artifact) could not use the microphone, and browser text-to-speech sounded robotic.

3. User stories

Id	As a candidate I want to…	So that…
VOICE-S1	answer out loud and have the app notice when I'm done	I never press a button between turns
VOICE-S2	hear the interviewer start talking quickly after I stop	it feels like a conversation
VOICE-S3	have my technical words transcribed right (Redis, Kafka, idempotency key)	the interviewer judges what I said, not what the recogniser heard
VOICE-S4	cut the interviewer off mid-sentence	I can correct course like in a real call
VOICE-S5	mute without losing what I'd said so far	I can fix it by typing
VOICE-S6	check my mic and speaker before joining	I don't find out mid-round that the mic is wrong
VOICE-S7	each interviewer to keep the same voice all round	personas feel like people
VOICE-S8	carry on typed if voice breaks	a round never dead-ends

4. The experience

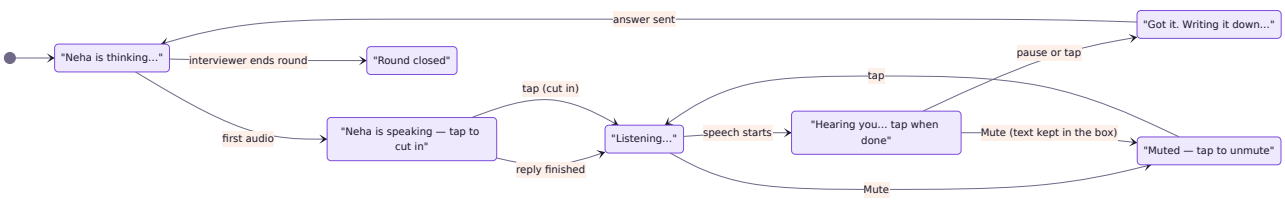
Get ready → room ready → live



The **room ready** checklist shows each step as *running*, *ok*, *warning*, *failed* or *off*. The round clock only starts when the room is ready, so loading time never counts against the candidate.

The orb: what the candidate sees during a turn

The microphone orb in the middle of the room is the one control that matters. Its label always says what is happening:

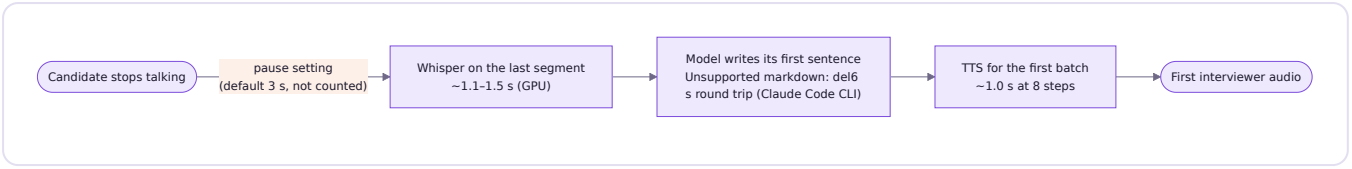


- **Pause to finish:** 1.8 s, 3 s (default), 5 s, or "Never — I tap the mic".
- **Live caption** shows the words as they are heard.
- **Waveforms** on both sides of the orb move with the mic or the interviewer's voice.

5. Requirements

Id	Requirement	Status
VOICE-R1	Speech in both directions uses VoiceStudio on `127.0.0.1:3900`. No audio leaves the machine.	Shipped
VOICE-R2	The answer ends after the chosen pause of silence, or on tap.	Shipped
VOICE-R3	Long answers are cut into segments at short pauses so transcription runs while the candidate talks.	Shipped
VOICE-R4	Each segment is re-transcribed by Whisper large-v3 on the GPU with a tech vocabulary prompt. Whisper's text is what's sent; Parakeet's is the fallback.	Shipped
VOICE-R5	Repeated phrases ("for the MQ for the MQ") are collapsed and common tech terms are fixed.	Shipped
VOICE-R6	Interviewer text is spoken sentence by sentence as it streams. Short sentences are merged (max 220 chars per call).	Shipped
VOICE-R7	Code blocks are never read aloud.	Shipped
VOICE-R8	Tapping the orb while the interviewer speaks stops the audio and opens the mic (barge-in).	Shipped
VOICE-R9	Mute cancels listening and puts what was heard so far in the typing box.	Shipped
VOICE-R10	Each panel persona has a fixed VoiceStudio voice profile, built once and reused.	Shipped
VOICE-R11	If VoiceStudio, the voice or the mic fails at start, the candidate can start the round typed.	Shipped
VOICE-R12	Mic or speaker can be switched in the room. An unplugged mic ends listening with a clear message.	Shipped
VOICE-R13	English only: the STT model is English-only Parakeet v2 (installed on demand); TTS always gets `language: English`.	Shipped
VOICE-R14	First interviewer audio < 2.5 s after the candidate stops talking.	Partial: being tuned (see below)

6. Latency budget



The pause setting itself (default 3 s) comes first and is not counted. Measured numbers live in `CLAUDE.md`: Whisper 1.1–1.5 s per answer on the GPU, TTS ~1.0 s for a short sentence at 8 steps, Groq TTFT ~0.4–0.5 s, Claude Code CLI ~6 s round trip on a warm process. With the CLI, the budget is not met yet; API providers come close.

7. Settings (VoiceStudio section)

Setting	Default	Notes
Spoken interviews on/off	On	Off = every round is typed.
VoiceStudio address	`http://127.0.0.1:3900`	
Voice	first profile	Used when the round has no panel persona. VoiceStudio's random "default" voice is never used.
Quality (steps)	8 = fast	16 balanced, 32 best.
Speed	1.0	Multiplied by the persona's own speed factor.
Pause that ends an answer	3 s	1.8 s / 3 s / 5 s / never.
Microphone, speaker	system default	Also switchable on Get ready and in the room.

8. Dependencies and risks

Risk	Effect	Mitigation
VoiceStudio cold start (~70 s to start, ~60 s first speech)	First round waits	Warm-up at app launch and at round start.
VoiceStudio unloads after ~15 min idle	Next call ~10 s	Warm-up line at round start when it hasn't spoken in 5 min.
VoiceStudio demotes a dictation model after one silent session	STT silently falls back to whisper-tiny	Onsite re-selects the model at round start and restores the user's pick.
No GPU	No Whisper pass	Parakeet text is used.

Risk	Effect	Mitigation
Echo from speakers into the mic	Interviewer hears itself	The mic is closed while the interviewer speaks, unless the candidate cuts in.